

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills		
Student Name:	Joanathan Packia Singh	Reg. No.:	192472229
Date:	05/08/2026	Duration:	60 minutes

Code of Conduct

I Joanathan Packia Singh(192472229) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Joanathan

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction
 - Problem Description
 - Scope of the System
 - Functional Requirements

- Non-Functional Requirements
 - Use Case Descriptions
 - Data Requirements
 - External Interface Requirements
 - System Features
5. **Identify Assumptions and Constraints**
- List the assumptions considered while developing the system.
 - Identify technical, operational, economic, legal, and time-related constraints affecting the project.
6. **Validate Requirement Completeness and Consistency**
- Verify whether all stakeholder requirements have been addressed.
 - Check for ambiguity, redundancy, inconsistency, and missing requirements.
 - Suggest improvements to enhance the quality and completeness of the requirement specification.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement	Clearly identifies realistic assumptions and constraints; validates requirements	Identifies most assumptions and constraints;	Limited assumptions/constraints;	Assumptions, constraints, and validation are

Validation (10%)	for completeness, consistency, and ambiguity with appropriate justification.	validation is adequate.	validation is superficial.	missing or incorrect.
-----------------------------	--	-------------------------	----------------------------	-----------------------

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25

SIMATS ENGINEERING
(SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1
Scenario-Based Requirement Analysis

Software Requirement Specification (SRS)

22. Fitness and Diet Tracker with Personalized Recommendations

Student Name	Joanathan Packia Singh	Reg. No.	192472229
Course Code	CSA1016	Duration	60 minutes
Course Title	Software Engineering	Weightage	20%
CO Assessed	CO2	Total Marks	25
Assessment	Assessment Tool 1 – Scenario-Based Requirement Analysis		
Date	05/08/26	Problem No.	22

Real-Time Tool Used for Requirement Analysis: Jira

1. Introduction

1.1 Purpose

This document presents a Scenario-Based Requirement Analysis and Software Requirement Specification (SRS) for the Fitness and Diet Tracker with Personalized Recommendations. The purpose of this document is to analyze the assigned problem statement, identify the stakeholders and business context, extract functional and non-functional requirements, model the system actors and use cases, and validate the completeness of the gathered requirements. This document is intended to guide the design and development phases of the proposed system.

1.2 Industry Context

The fitness and wellness industry has experienced significant growth due to increasing awareness of healthy lifestyles and preventive healthcare. People increasingly seek digital solutions to monitor physical activities, calorie intake, and overall well-being. Fitness centers, nutritionists, and healthcare providers require integrated platforms to offer personalized health guidance, while wearable devices and mobile applications generate valuable health data that can be analyzed to produce individualized recommendations.

1.3 Document Scope

This SRS covers the requirement analysis phase only. It captures the problem description, system scope, functional and non-functional requirements, actors and use cases, data and interface requirements, system features, assumptions, constraints, and requirement validation for the Fitness and Diet Tracker application.

1.4 Tools Used for Requirement Analysis

As instructed, the real-time project management tool Jira was used to organize and track the requirement analysis activities for this assignment, including epics, user stories, and the requirement backlog. Screenshots of the Jira workspace used for this analysis are attached in Section 14 of this document.

1.5 Requirement Analysis Process

The requirement analysis for this project followed the structured process illustrated below, moving from scenario study through to validation and real-time tracking in Jira.



Figure 1.1: Requirement Analysis Process Flow

2. Problem Description

Design and develop a Fitness and Diet Tracker that enables users to monitor daily physical activities, calorie intake, and nutritional habits. The platform must collect user health data, including age, weight, BMI, and fitness goals, in order to generate personalized recommendations. It should provide customized workout schedules and diet plans based on individual preferences and health conditions.

The system must track user progress using interactive dashboards and visual reports. Integration with wearable devices should allow automatic synchronization of health metrics such as step count, heart rate, and sleep patterns. The application should send timely reminders for workouts, hydration, and meals while ensuring the privacy and security of sensitive health data. The overall solution must offer an intuitive interface that encourages consistent, healthy lifestyle management.

2.1 Existing Challenges

- Many existing fitness applications lack comprehensive personalization based on individual health data.
- Limited integration between wearable devices and diet-tracking modules.
- Poor progress visualization, making it difficult for users to stay motivated.
- Inconsistent reminder systems that fail to encourage habit formation.
- Weak data privacy safeguards for sensitive personal health information.

2.2 System Objectives

1. Develop a personalized fitness and diet management platform.
2. Track daily physical activities and nutritional intake.
3. Generate AI-based workout and diet recommendations.
4. Integrate wearable device data for continuous health monitoring.
5. Visualize user progress through reports and dashboards.
6. Send reminders for exercise, meals, and hydration.
7. Ensure secure storage of personal health information.
8. Promote healthy lifestyle habits through continuous monitoring.

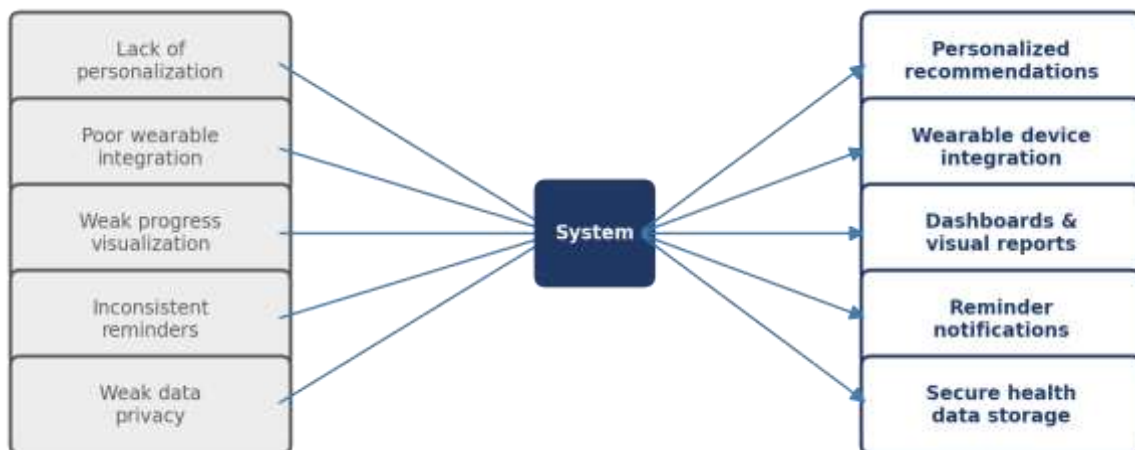


Figure 2.1: Mapping of Identified Challenges to System Objectives

3. Scope of the System

The Fitness and Diet Tracker is a cross-platform application (web and mobile) that allows individual users to record their health profile, track daily activity and nutrition, receive personalized workout and diet recommendations, and monitor long-term progress. The system will synchronize with common wearable devices and third-party fitness APIs to automatically collect activity data, reducing manual entry.

The system is intended for individual end users, with a secondary interface for nutritionists or fitness coaches to review client progress where applicable. The scope of this analysis is limited to the core tracking, recommendation, and reminder functionality; it excludes hardware manufacturing of wearable devices and does not provide medical diagnosis.

3.1 In Scope

- User registration, health profiling, and goal setting.
- Daily activity logging and calorie intake tracking.
- Personalized AI-based workout and diet plan generation.
- Wearable device data synchronization.
- Progress dashboards, visual reports, and reminders.
- Secure storage and management of personal health data.

3.2 Out of Scope

- Manufacturing or firmware development of wearable hardware.
- Clinical diagnosis or prescription of medical treatment.
- Direct billing or insurance-claim processing.



Figure 3.1: System Scope Boundary – In Scope vs. Out of Scope

4. Stakeholder and Business Domain Analysis

The Fitness and Diet Tracker operates within the health, wellness, and fitness technology domain. The following stakeholders were identified through analysis of the problem statement and industry context.

4.1 Primary Stakeholders

- End Users / Fitness Enthusiasts – individuals tracking their fitness and diet.
- Nutritionists / Fitness Coaches – professionals reviewing user progress and adjusting plans.
- System Administrators – responsible for platform operation and data governance.

4.2 Secondary Stakeholders

- Wearable Device Manufacturers – provide data feeds through APIs.
- Healthcare Providers – may consume aggregated wellness data with user consent.
- Application Development and Support Team – build and maintain the platform.

4.3 Business Value

By bridging the personalization gap present in existing fitness applications, the platform is expected to increase user engagement, improve adherence to fitness and nutrition goals, and generate a scalable data-driven service that benefits both individual users and affiliated wellness professionals.

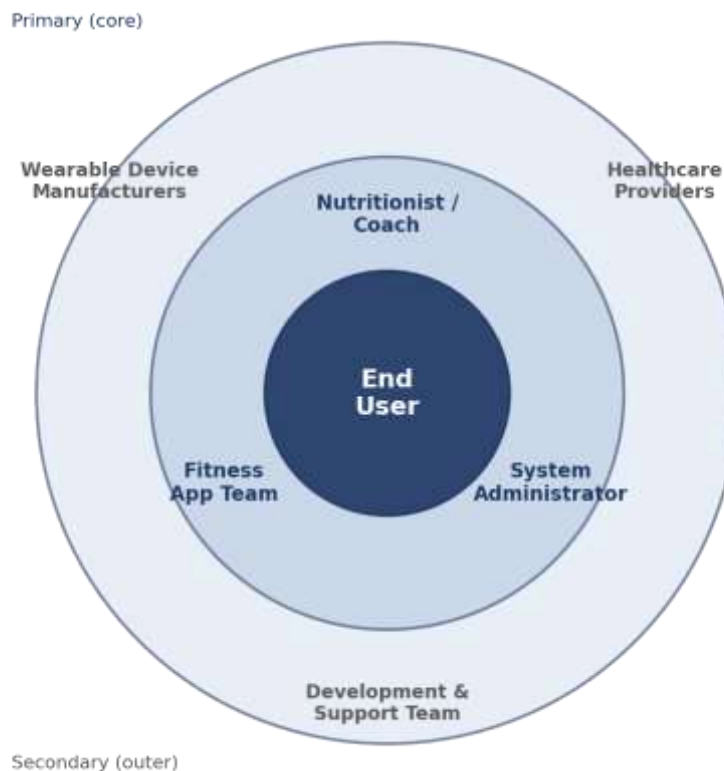


Figure 4.1: Stakeholder Map – Primary and Secondary Stakeholders

5. Functional Requirements

The following functional requirements define the capabilities the system must provide.

1. FR1: The system shall allow users to register and create a personal health profile including age, gender, height, weight, and fitness goals.
2. FR2: The system shall automatically calculate BMI and recommended calorie targets from the user profile.
3. FR3: The system shall allow users to log daily physical activities such as exercises, steps, and duration.
4. FR4: The system shall allow users to log daily meals and calculate nutritional intake.
5. FR5: The system shall generate personalized workout schedules based on user goals and health conditions.
6. FR6: The system shall generate personalized diet plans based on user preferences and nutritional needs.
7. FR7: The system shall synchronize data automatically from connected wearable devices.
8. FR8: The system shall display progress through interactive dashboards and visual reports (charts, graphs, trends).
9. FR9: The system shall send reminders and notifications for workouts, hydration, and meals.
10. FR10: The system shall allow users to set, update, and track fitness goals over time.
11. FR11: The system shall allow nutritionists or coaches to view and comment on assigned users' progress.
12. FR12: The system shall allow users to export or share progress reports.
13. FR13: The system shall provide secure user authentication and authorization.

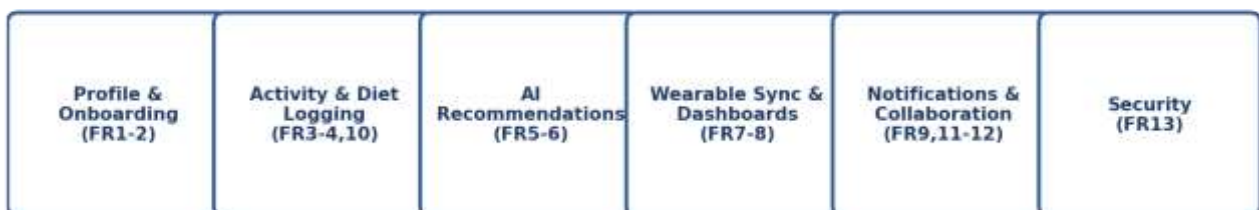


Figure 5.1: Functional Requirement Groupings

6. Non-Functional Requirements

6.1 Performance

The system shall process and reflect activity or nutrition log updates within 2 seconds under normal load, and generate AI-based recommendations within 5 seconds of a request.

6.2 Security

The system shall encrypt personal health information both in transit and at rest, and shall comply with applicable data protection regulations governing sensitive health data.

6.3 Reliability

The system shall maintain 99.5% uptime and shall not lose logged user data in the event of a service interruption.

6.4 Usability

The system shall provide an intuitive, minimal-effort interface that allows a new user to log a meal or activity within three taps or clicks.

6.5 Scalability

The system architecture shall support horizontal scaling to accommodate growth in the number of concurrent users and wearable device data streams.

6.6 Maintainability

The system shall be built using modular components to allow independent updates to the recommendation engine, tracking module, and notification service.

6.7 Availability

Core tracking and logging features shall remain available even under intermittent network connectivity, using local caching with later synchronization.

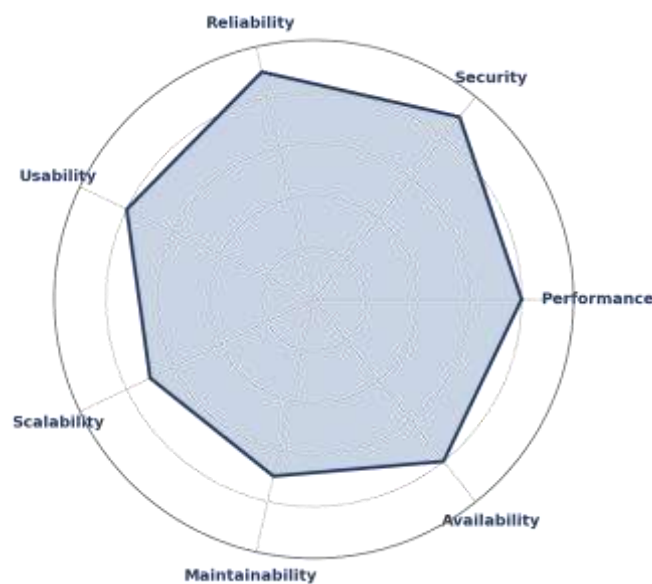


Figure 6.1: Relative Priority of Non-Functional Quality Attributes

7. Actors and Use Cases

7.1 Primary Actors

- End User – registers, logs activity and diet, views recommendations and progress.
- Nutritionist / Fitness Coach – reviews assigned users' data and adjusts recommendations.

7.2 Secondary Actors

- Wearable Device / Fitness API – supplies automatic activity and health data.
- Notification Service – triggers reminders for workouts, hydration, and meals.
- System Administrator – manages users, data governance, and platform configuration.

7.3 Major Use Cases

1. Register and Create Health Profile
2. Log Physical Activity
3. Log Meal and Calorie Intake
4. Generate Personalized Workout Plan
5. Generate Personalized Diet Plan
6. Synchronize Wearable Device Data
7. View Progress Dashboard and Reports
8. Receive Reminders and Notifications
9. Review Client Progress (Nutritionist / Coach)
10. Manage Users and Data (Administrator)

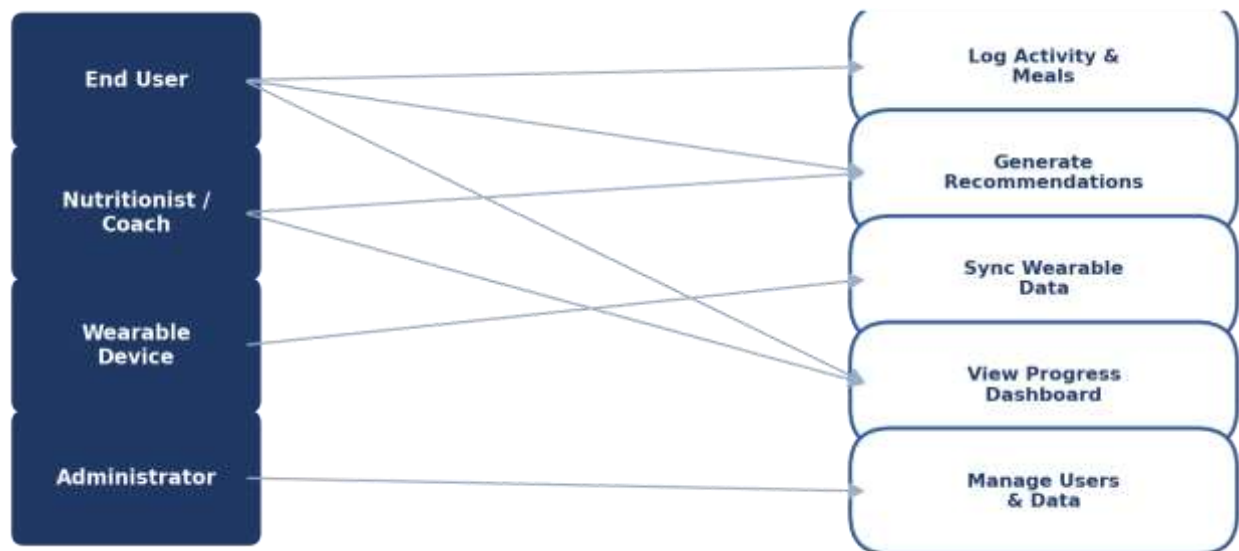


Figure 7.1: Actor – Use Case Interaction Map

8. Use Case Diagram

The use case diagram below represents the interactions between the identified actors and the Fitness and Diet Tracker system, covering registration, activity and diet logging, personalized recommendation generation, wearable synchronization, progress visualization, and reminder notifications.

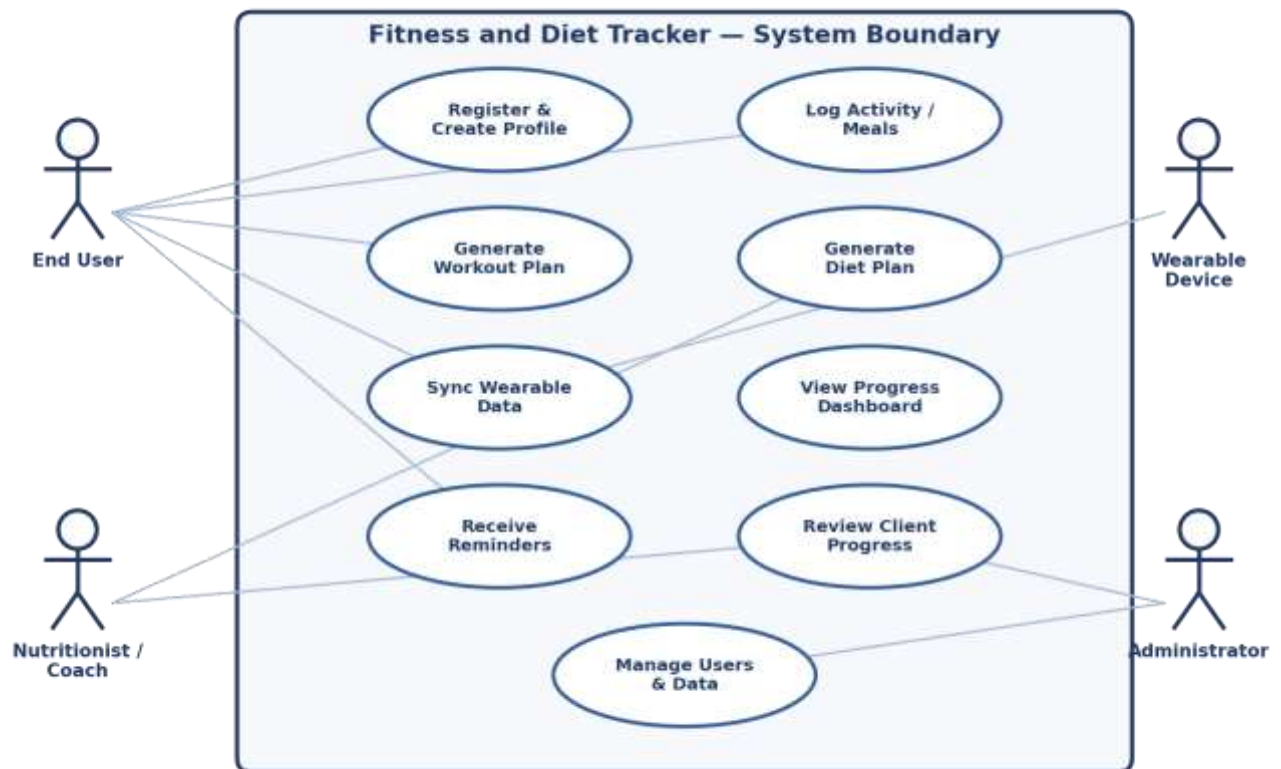


Figure 8.1: UML Use Case Diagram for the Fitness and Diet Tracker System

The diagram uses standard UML notation, showing association relationships between each actor and their respective use cases. Relationships such as 'Generate Diet Plan' draw on data captured through 'Sync Wearable Data' and 'Log Activity / Meals', reflecting an extend-style dependency between these use cases.

9. Data Requirements

The system must capture, store, and process the following categories of data to support personalized recommendations and progress tracking.

9.1 User Profile Data

- Name, age, gender, height, weight, and contact details.
- Fitness goals (weight loss, muscle gain, endurance, general wellness).
- Existing health conditions or dietary restrictions.

9.2 Activity and Nutrition Data

- Daily exercise logs (type, duration, intensity, calories burned).
- Meal logs (food items, portion size, calorie and macronutrient breakdown).
- Hydration logs and sleep data synchronized from wearables.

9.3 System-Generated Data

- Calculated BMI and calorie targets.
- Personalized workout and diet plan records.
- Progress history used for trend visualization and report generation.

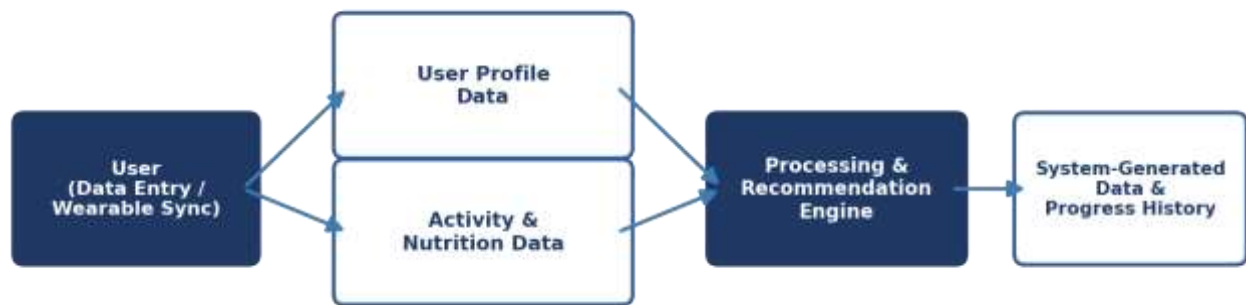


Figure 9.1: Data Flow – From User Input to System-Generated Records

10. External Interface Requirements

10.1 User Interfaces

A responsive mobile and web dashboard providing profile management, activity and meal logging forms, recommendation views, progress charts, and a notification center.

10.2 Hardware Interfaces

Integration with wearable fitness devices (smartwatches, fitness bands) capable of transmitting step count, heart rate, and sleep data via Bluetooth or a manufacturer API.

10.3 Software Interfaces

REST APIs for wearable device synchronization, a recommendation engine service for AI-based plan generation, and a notification/messaging service for reminders.

10.4 Communication Interfaces

HTTPS for all client-server communication, with push notification protocols used to deliver reminders to mobile devices.

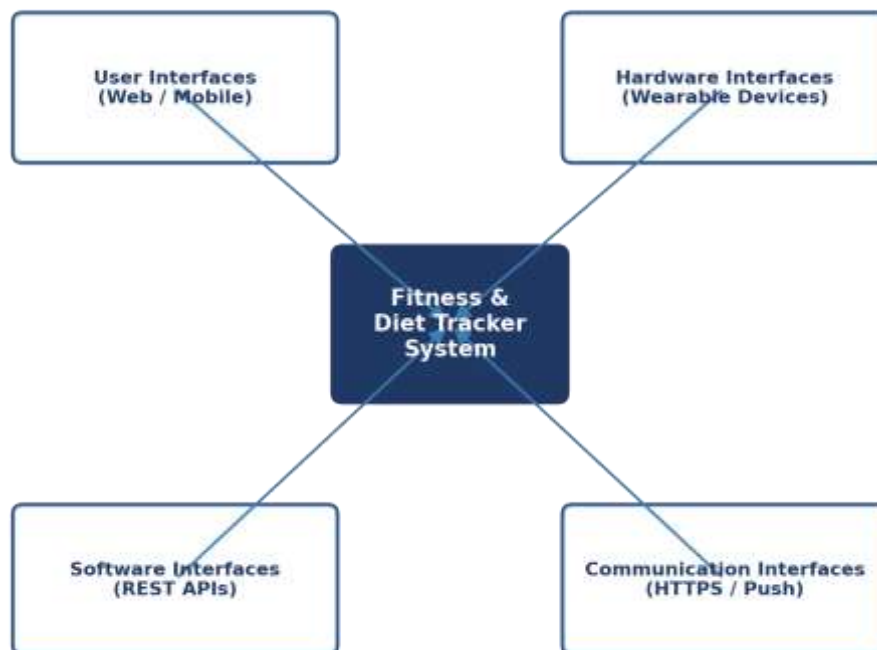


Figure 10.1: External Interface Categories Around the Core System

11. System Features

11.1 Personalized Recommendation Engine

Generates AI-based workout and diet plans by analyzing the user's profile, goals, and logged activity and nutrition history.

11.2 Activity and Nutrition Logging

Allows users to manually log or automatically sync physical activities and meals, with real-time calorie and nutrient calculations.

11.3 Progress Dashboard

Displays interactive charts and visual reports summarizing weight trends, calorie balance, and workout consistency over time.

11.4 Wearable Integration

Automatically synchronizes step count, heart rate, sleep, and calorie data from connected wearable devices.

11.5 Reminders and Notifications

Sends configurable reminders for scheduled workouts, hydration intervals, and meal times to encourage consistency.

11.6 Data Privacy and Security

Provides encrypted storage, access controls, and consent management for all personal health information collected by the system.

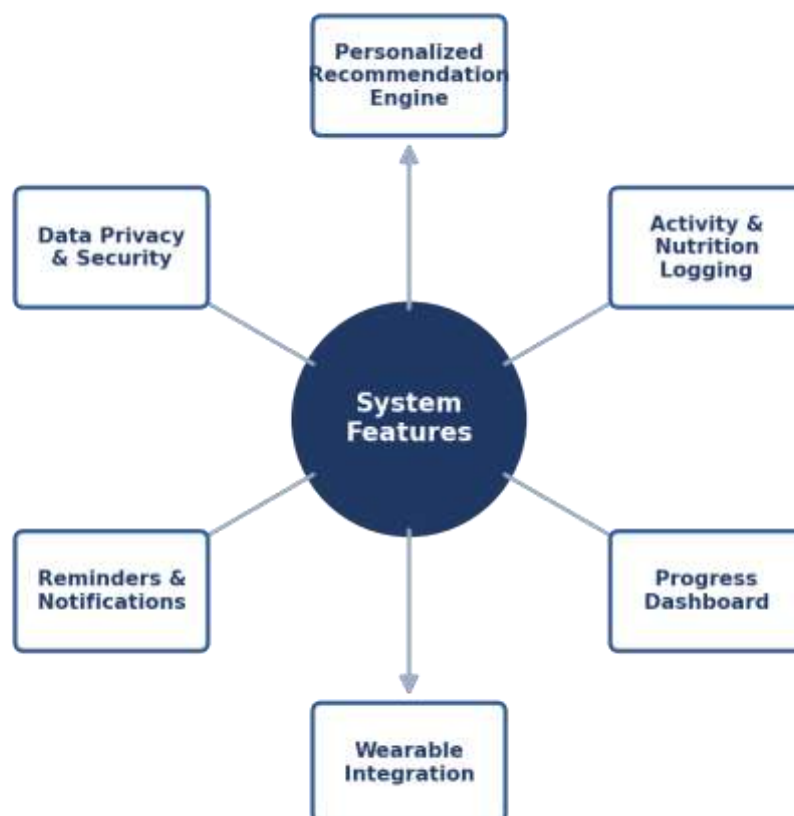


Figure 11.1: Core System Features Overview

12. Assumptions and Constraints

12.1 Assumptions

- Users have access to a smartphone or computer with internet connectivity.
- Wearable devices used by the target audience expose data through a standard API.
- Users provide accurate profile information for meaningful personalization.
- Nutritionists and coaches have basic digital literacy to use the review interface.

12.2 Technical Constraints

- The recommendation engine depends on the availability and accuracy of third-party wearable APIs.
- Offline logging requires local caching and later synchronization, adding implementation complexity.

12.3 Operational Constraints

- The system must be maintained continuously to keep recommendation models current.

12.4 Economic Constraints

- Development and cloud hosting costs must remain within the allocated project budget.

12.5 Legal Constraints

- The system must comply with health-data privacy regulations applicable to the target region.

12.6 Time-Related Constraints

- The requirement analysis and subsequent design phases must be completed within the academic assessment schedule.

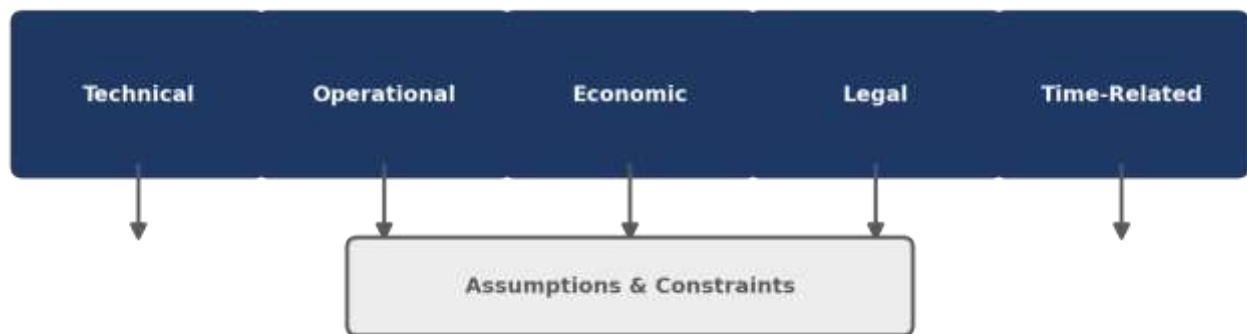


Figure 12.1: Categories of Constraints Affecting the System

13. Requirement Validation and Consistency Check

13.1 Completeness Check

All objectives listed in the problem statement — activity tracking, calorie tracking, personalized recommendations, wearable integration, progress visualization, and reminders — are addressed by corresponding functional requirements (FR1–FR13), confirming stakeholder requirements have been covered.

13.2 Ambiguity Check

Requirements were reviewed to remove vague terms; for example, ‘fast performance’ was refined into the measurable Non-Functional Requirement in Section 6.1 specifying response times in seconds.

13.3 Redundancy Check

Overlapping requirements between activity logging and wearable synchronization were consolidated so that manual and automatic data entry are treated as complementary paths into the same activity log, avoiding duplicate requirements.

13.4 Consistency Check

Functional requirements were cross-checked against the identified use cases in Section 7 to confirm that every use case maps to at least one functional requirement, and no requirement contradicts another.

13.5 Suggested Improvements

- Add a requirement for offline-first data capture to strengthen availability.
- Add explicit consent-management requirements for data shared with nutritionists.
- Define measurable accuracy targets for the AI recommendation engine in a future iteration.



Figure 13.1: Requirement Validation Pipeline

14. Real-Time Tool Usage – Jira for Requirement Analysis

As per the instructions provided for this assessment, Jira was used as the real-time software requirement analysis and tracking tool for the Fitness and Diet Tracker project. The requirement analysis activities, including epics, user stories, and backlog prioritization, were organized within a Jira project. The screenshots below capture the Jira workspace used during this analysis.

14.1 Jira Project Backlog

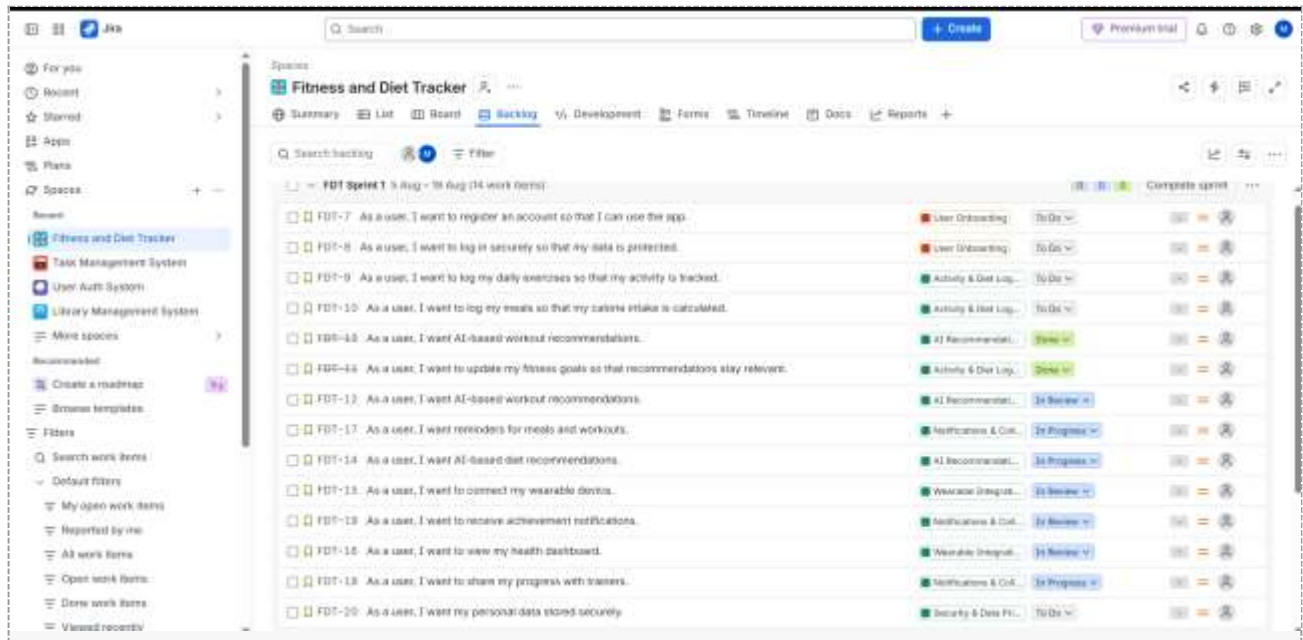


Figure 14.1: Jira Backlog View

14.2 Jira Board – Requirement Tracking

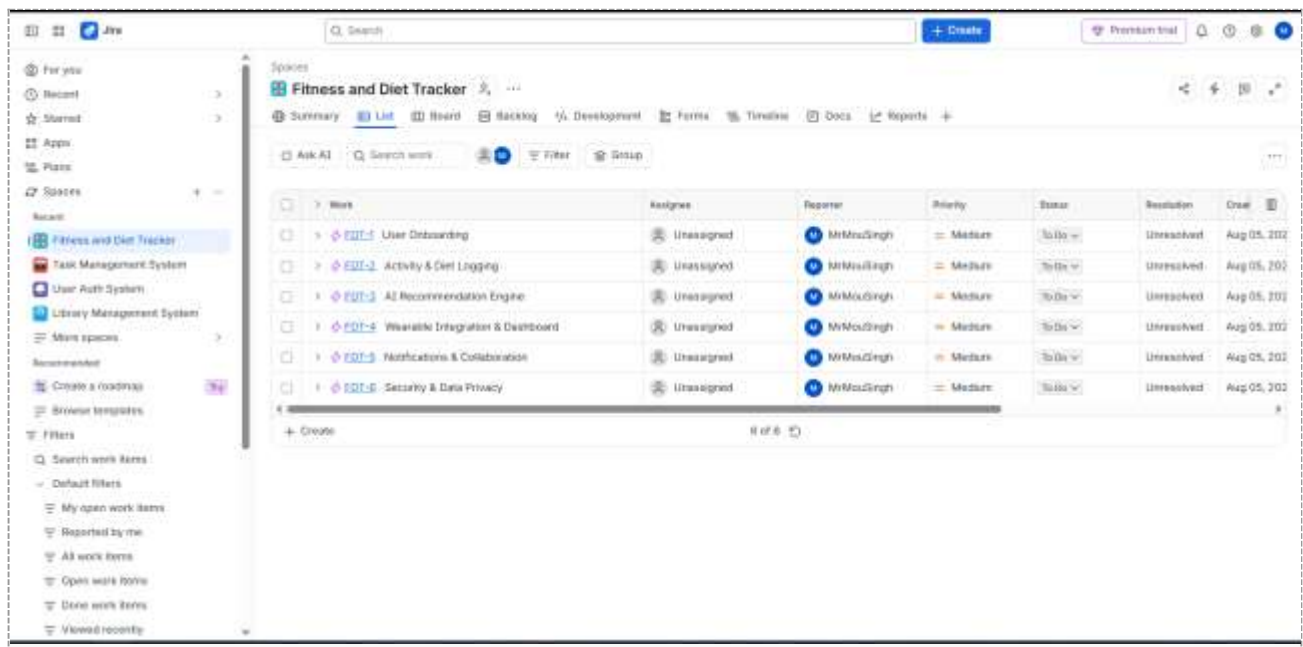


Figure 14.2: Jira Board View

14.3 Jira Board

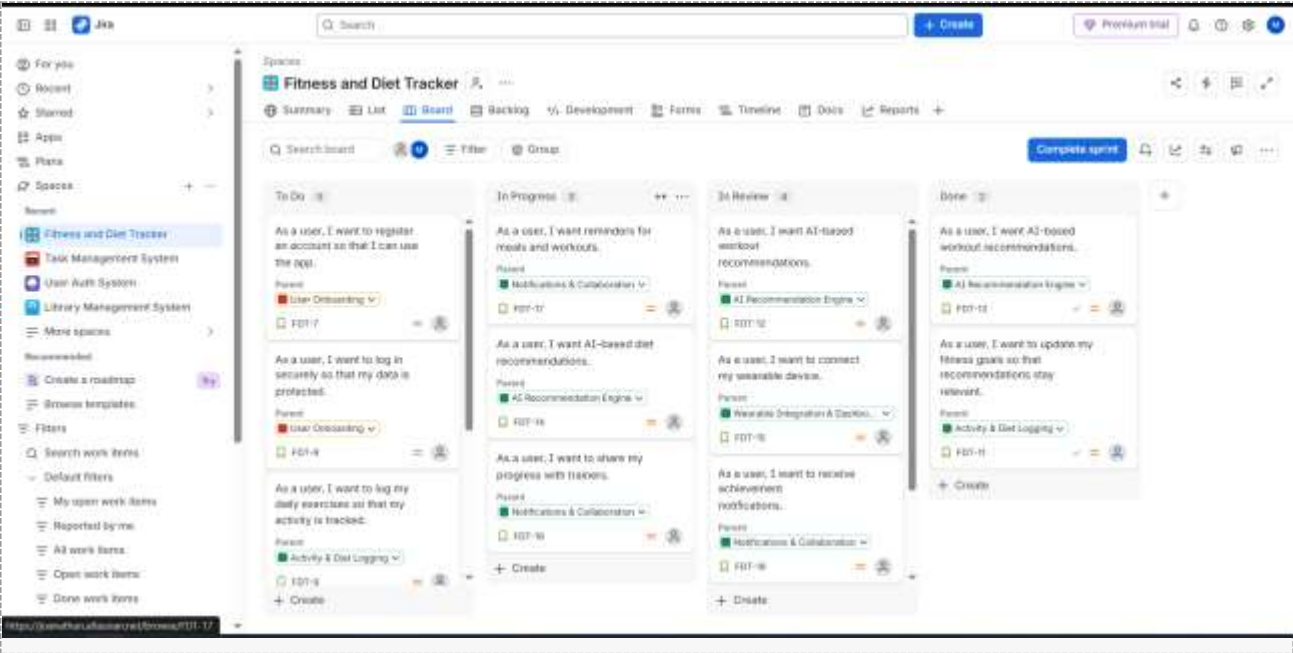


Figure 14.3: Jira User Story Detail View

14.3 Jira User Story Detail

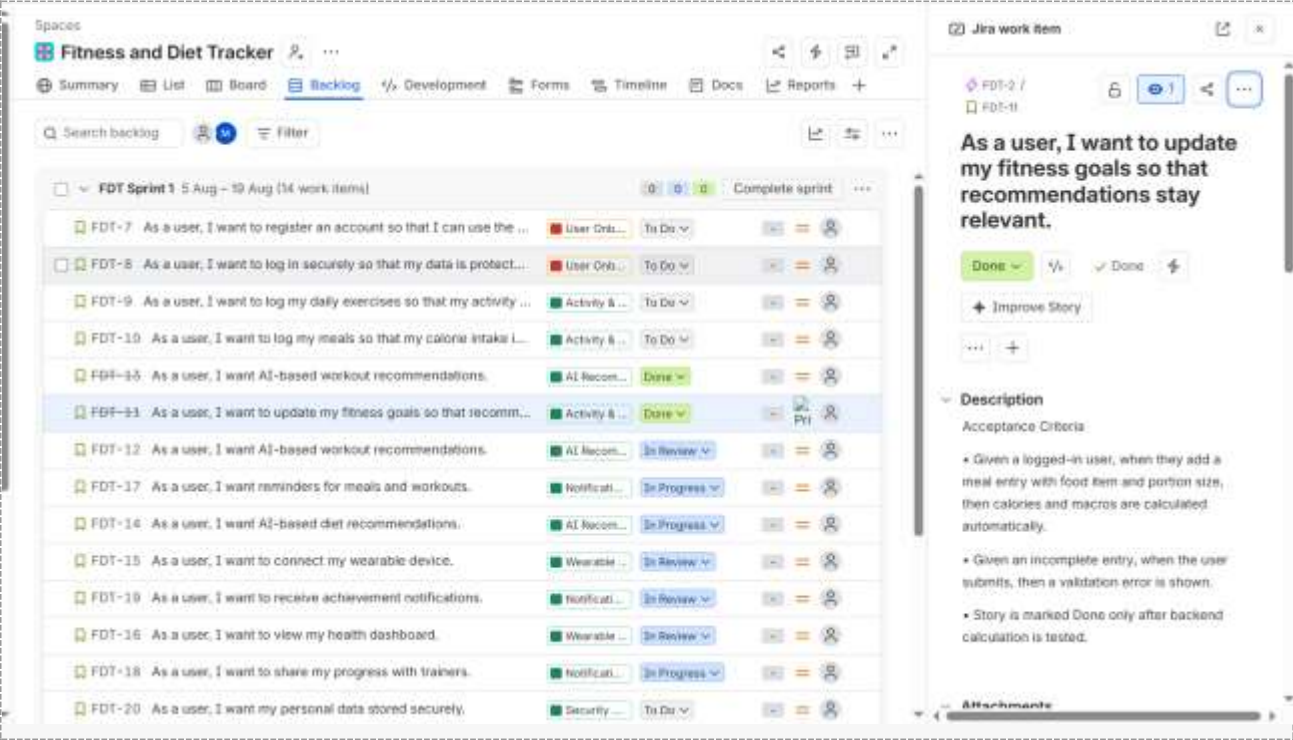


Figure 14.3: Jira User Story Detail View

14.3 Jira Summary View

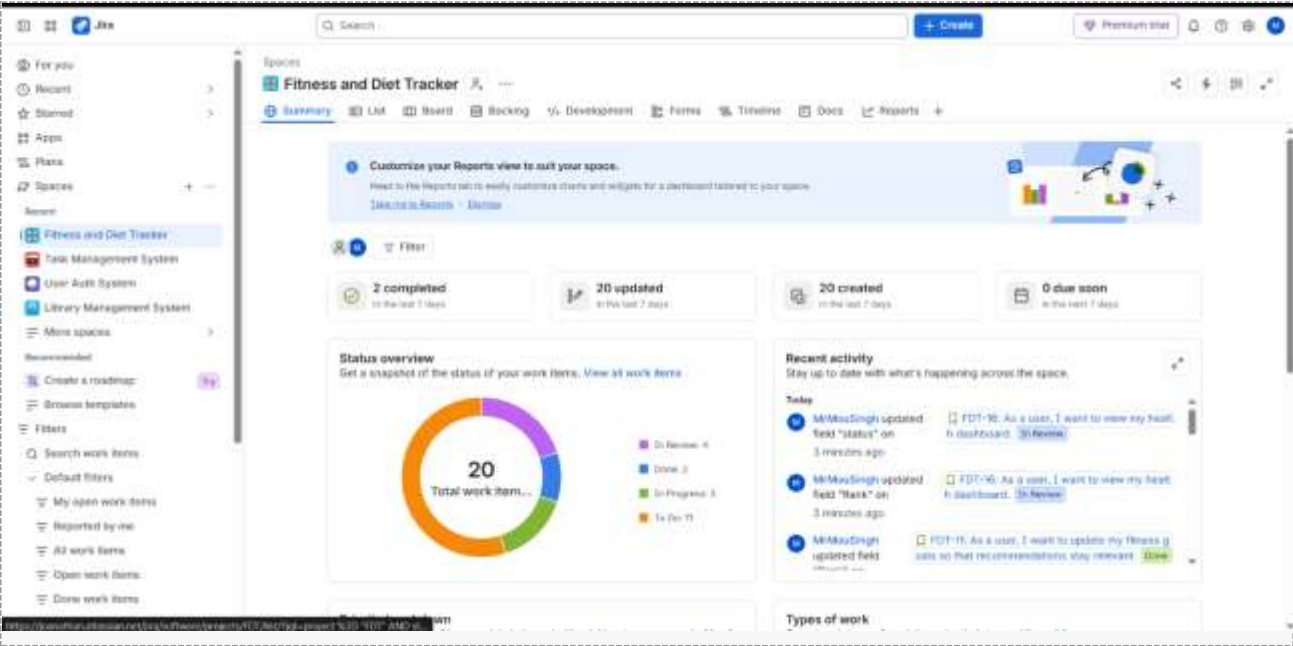


Figure 14.3: Jira User Story Detail View

15. Conclusion

This Scenario-Based Requirement Analysis has examined the Fitness and Diet Tracker with Personalized Recommendations problem statement in depth, identifying the relevant stakeholders, business domain, functional and non-functional requirements, actors, use cases, data and interface requirements, system features, assumptions, and constraints.

The requirement set was validated for completeness, consistency, and freedom from ambiguity, and real-time tracking of the analysis was performed using Jira as instructed. This document provides a solid foundation for the subsequent design and development phases of the proposed system.



Figure 15.1: Document Roadmap – Summary of Sections Covered